Thinkerbell Labs Private Limited
CIN: U72900KA2016PTC094046
4th Floor, JK Towers, 46th cross, Marenahalli Road, 8th Block, Jayanagar, Bengaluru - 560070

Safe device settings update

60-minute preparation cap | AI and documentation allowed

[Certain] This handout defines a simulated exercise with synthetic data. The task and assessment rules below are proposals for the hiring team.

Your task

Implement **apply_update(current, actor, request)** in a language you use confidently. Return a result with an HTTP-style status and either the updated record or an error code. Keep it a small, runnable module; server, database and UI setup are outside this 60-minute task.

Scenario: a staff member edits the language and volume settings of a learning device. Another staff member may have edited the same record since the form loaded.

Inputs

```
current = {"deviceId":"D01", "siteId":"S01", "version":3,
           "settings":{"language":"en", "volume":2}}
actor   = {"siteId":"S01", "role":"editor"}
request = {"deviceId":"D01", "expectedVersion":3,
           "settings":{"volume":4}}
```

Treat **current** as a valid saved record and **actor** as trusted, server-authenticated identity. Treat **request** as untrusted parsed JSON. The settings payload is a partial update.

Required behavior - evaluate in this order

Check	Required result
1. Permission	Role must be editor and actor.siteId must match current.siteId; otherwise 403 / FORBIDDEN.
2. Request structure	Require exactly deviceId, expectedVersion and settings. deviceId is a non-empty string; expectedVersion is a positive integer (not a boolean); settings is a non-empty object. Otherwise 422 / INVALID_INPUT.
3. Settings validation	Allow only language (en, hi or kn) and volume (integer 0-5, not a boolean). Reject unknown keys and invalid values with 422 / INVALID_INPUT.
4. Identity and version	Wrong deviceId: 404 / NOT_FOUND. Different expectedVersion: 409 / CONFLICT.
5. Success	Return 200 with a NEW record: merge only supplied settings and increment version once. Never mutate current, actor or request, on success or failure.

Minimum work

- Implement the function and at least two automated tests: one successful partial update and one rejected update that leaves input state unchanged.
- In at most eight README bullets, describe a UI save flow: labels/keyboard use, pending state, validation error, conflict recovery and how you avoid showing an unsaved value as saved.

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Submission and live review

Examples to check your interpretation

For the sample input, return 200 with version 4 and settings {language: "en", volume: 4}. The original current object must still have version 3 and volume 2.

Change expectedVersion to 2: return 409 / CONFLICT with no state change. Change role to viewer: return 403 / FORBIDDEN. These examples do not replace the other rules.

What to submit

- A ZIP or repository link containing your code, tests and a short README. Use synthetic data only; share code you have permission to share.
- README: one run command, one test command, time spent, unfinished work, one known limitation, and a short note on your design choices.
- List AI tools and outside code used, and what you checked yourself. A brief summary is enough; do not submit private chat histories. AI use is permitted and is not a penalty.
- Stop after 60 minutes. Submit partial work with clear limits. If setup blocks progress, record the blocker and ask the hiring contact for an equivalent live exercise.

Send your submission through the channel used for the invitation. Agree the deadline with the hiring contact before starting.

The submission supplies material for the live session; it is not an ability score by itself. You may use AI and docs live. Explain your prediction before running a change, then verify the result.